

MEHADI HASAN

SENIOR SOFTWARE ENGINEER · TECH LEAD · AI / LLM SOLUTIONS ARCHITECT

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14+ Years · Government · Fintech · Healthcare · Enterprise

Production GenAI & Agentic AI · Mission-Critical Systems ·

Multi-Team Leadership

PROFESSIONAL SUMMARY

Results-driven technologist with **14+ years** delivering mission-critical systems across **government, fintech, healthcare** and **enterprise** domains. Architect of production **Generative AI & Agentic AI** platforms – **LLM integration** (OpenAI, Anthropic Claude, Google Gemini, Ollama), **RAG pipelines**, **LangChain / LangGraph** orchestration, and **Vector Databases** (Pinecone, Weaviate, Qdrant, Chroma, FAISS, Milvus, pgvector) on **AWS Bedrock** and **Azure OpenAI**. Power user of **AI pair programming** (Claude Code, GitHub Copilot, Gemini Code Assist, OpenAI Codex). Deep expertise in **microservices**, **system design (HLD/LLD)**, **event-driven architecture** (Kafka, RabbitMQ), **n8n workflow automation**, and **MLOps / LLMops**. Currently leading 3 Malaysia Government transport & infrastructure programmes (LRT3, RTS, Smart Car Parking) as **Senior Engineer & Lead** at GIS. Pursuing **PhD in Computer Science**; MSc thesis on AI-driven medical assistant systems.

CORE COMPETENCIES

LLM Integration Agentic AI RAG Vector Search LangChain LangGraph LlamaIndex AWS Bedrock Azure OpenAI Vertex AI
MCP Servers Prompt Engineering LLM Evals MLOps / LLMops n8n HLD/LLD Microservices Event-Driven Kafka RabbitMQ
Real-time Cloud Native CI/CD OWASP Top 10 AI Security Open Source Team Leadership

TECHNICAL SKILLS

AI / GenAI Stack: OpenAI, Claude, Gemini, Grok, Mistral; Ollama, vLLM, llama.cpp; LangChain, LangGraph, LlamaIndex, Semantic Kernel; RAG, MCP, Function Calling, ReAct, AutoGen; AWS Bedrock, Azure OpenAI, Vertex AI; LangSmith, Langfuse, RAGAS, Promptfoo

Vector Databases: Pinecone, Weaviate, Qdrant, Chroma, FAISS, Milvus, pgvector

Programming Languages: Python, JavaScript/TypeScript, Java, C/C++, C#, PHP, Dart, Kotlin, Go

Frontend: React.js, Vue.js, Flutter, WPF, HTML5/CSS3

Backend: Node.js, NestJS, ASP.NET Core, Laravel, Spring Boot, FastAPI; REST, GraphQL, gRPC, SOAP

AI Pair Programming: Claude Code, GitHub Copilot, Gemini Code Assist, OpenAI Codex

Architecture: System Design (HLD/LLD), Microservices, Event-Driven, Domain-Driven Design, SOLID

Automation: n8n, Workflow Orchestration, AI-Driven Architecture, API Integration Patterns

Databases: PostgreSQL, MySQL, MongoDB, Redis, MSSQL, Oracle, SQLite, Firebase

Cloud / DevOps: AWS (Bedrock, EC2, S3, Lambda), Azure (OpenAI), GCP (Vertex AI), Docker, Kubernetes, CI/CD, VMware ESXi, Grafana

Messaging & Real-time: Kafka, RabbitMQ, WebSocket, MQTT, SignalR, JMS

Methodology & Tools: Agile/Scrum, TDD, Redmine, Jira, Git/Bitbucket, Jenkins, Neovim, LaTeX, Overleaf

Custom-Built Tools: MetaSig, KafkaTUI, RabbitMQTUI, IPScanner (custom network mapping & visualization), custom ticket monitoring

Networking & VPN: Tailscale, OpenVPN, WireGuard, multicast tracing, SignalR

PROFESSIONAL EXPERIENCE

Senior Software Engineer & Lead

Sep 2024 – Present

GIS – Government Projects: LRT3, RTS, Smart Car Parking · Malaysia

- ▶ Lead engineer across **3 simultaneous Malaysia Government programmes**; own end-to-end HLD/LLD for **4-team, 18-engineer** organisation
- ▶ **LRT3:** C# .NET + PostgreSQL signalling backend processing **>2M events/day**; cut alarm noise **~60%** via custom IP Scanner with multicast tracing
- ▶ **Smart Car Parking (TNG/Genting):** Flutter/Dart + NestJS + AI OCR plate detection; reduced gate latency from **4s** → **800ms** on integrated TNG/Mifare payment flow
- ▶ **MyDesk:** Custom AnyDesk-alternative unifying remote control of all 3 programmes; eliminated 3 commercial-licence costs (**~RM 60k/yr saved**)
- ▶ Introduced **LLM-assisted log triage** (Claude/OpenAI) + **n8n** ops pipelines; cut mean-time-to-diagnose by **~40%**
- ▶ Built **Kafka TUI**, **RabbitMQ TUI** now standard diagnostic for LRT3 ops team; SCADA via JMS; CISCO VMware ESXi; **Grafana** monitoring

Engineering Lead (Contract, Remote)

May 2025 – Oct 2025

Slotgator iGaming Project · Romania

- ▶ Delivered iGaming platform on **AWS** + **Kubernetes** sustaining **~5k concurrent sessions**; zero P0 incidents over 6-month engagement
- ▶ Drove **Kafka** streaming + microservices, AES-256 payment hardening; led 7-engineer distributed team across 4 time zones

- ▶ Shipped **4 hybrid iOS/Android apps** reaching **>100k installs**; managed **AWS EC2**, Tencent & Alibaba Cloud infra
- ▶ Integrated 3 payment rails (Decoin, Alipay, UnionPay) with AES-256 + real-time WebSocket; CI/CD cut release cycle from **2 weeks** → **2 days**

Project Manager / Team Lead
Creative IT Institute · Chittagong

Dec 2022 – Oct 2023

- ▶ Led **12-engineer** cross-functional team across 5 client products using Agile/Scrum; delivered **100% on-schedule** release rate
- ▶ Designed PostgreSQL/MySQL schemas powering **~30k DAU**; standardised Git, VPS & Nginx operating procedures; mentored **15+** junior engineers

Team Development Lead & Sr. Full Stack
Datahostbd · Dhaka

Mar 2022 – Oct 2022

- ▶ Mentored **8 junior engineers**; authored microservices training manual still in use; architected microservices with API gateway + service discovery (SOLID), reducing cross-service latency **~35%** vs. monolith baseline

Senior Consultant & Full Stack Developer
Creative IT Institute · Chittagong

Mar 2020 – Jan 2022

- ▶ Hardened 6 production codebases through code review + unit-test rollout; reduced post-release defects **~50% YoY**; coordinated deployments serving **~8k students**

Chief Consultant & Engineer
TRZ · Gazipur

Jan 2017 – Aug 2019

- ▶ Built end-to-end ERP for garments industry from scratch; RFID-integrated tracking across **4 factory floors**, **600+** daily operators
- ▶ Automated scheduled power management saving **~18%** electricity cost; BI model in Pandas/NumPy/Matplotlib adopted by exec leadership for daily KPI review

CEO & Chief Consultant
Orimpo Inc. Ltd · Dhaka

Jul 2011 – Nov 2016

- ▶ Founded & ran consultancy for 5+ years; delivered **40+ enterprise projects** on J2EE/JSP/Tomcat/J2ME across government, fintech, retail
- ▶ Led **bKash mobile payment prototyping** with Robi Axiata (Java) – foundational work for Bangladesh's largest mobile-money platform
- ▶ Directed engineering team; conducted security & network penetration assessments for govt & enterprise clients

SELECTED PROJECTS & OPEN-SOURCE

- ▶ **MetaSig – Cross-Platform Remote Device Management Platform** (2025 – Present) – Production fleet-management TUI for Android, iOS, macOS, Windows, Linux: WebSocket signalling, remote shell, file browser, live screen preview, APK packaging & signing pipeline. Manages **>200 endpoints** across 3 Malaysia Gov programmes from a single terminal. *Stack*: Node.js, neo-blessed, WebSocket, Android agent
- ▶ **GPT-5.3 Agent Bridge + Pentest-AI Red-Team Deployment** (2025 – 2026) – Built OpenAI-compatible HTTP bridge wrapping ChatGPT GPT-5.3 via Playwright; integrated as multi-model agent inside **opencode** TUI for agentic dev workflows. *Separately*: authorised internal red-team deployment of upstream open-source Pentest-AI (**ptai**, original by 0xSteph) for OWASP Top 10 exercises via MCP – *integration role, not authorship*
- ▶ **Smart Car Parking System (TNG / Genting)** (Sep 2024 – Present) – AI OCR plate detection + C/C++ hardware gate control + TNG/Mifare payment integration; cut gate-pass latency from **4s** → **800ms**. *Stack*: Flutter/Dart, NestJS, C# WinForms, Python OCR, PostgreSQL
- ▶ **LRT3 / RTS Signalling + KafkaTUI / RabbitMQ-TUI** (Sep 2024 – Present) – Mission-critical Malaysia Gov. transport signalling processing **>2M events/day** (C#/.NET, PHP, Java, Grafana); cut alarm noise **~60%**. KafkaTUI / RabbitMQ-TUI now standard diagnostic tools for LRT3 ops team
- ▶ **Medical AI Assistant (MSc Thesis) + AI Doctor Triage Platform** (2024 – 2025) – Healthcare AI with Claude / OpenAI & self-hosted Ollama; **RAG** over clinical KB (LangChain, LangGraph, Pinecone/Qdrant/pgvector, FastAPI, HIPAA-aware). Companion **AI Doctor** triage app: Gemini-powered, React Native mobile + Node.js backend + Prisma, Dockerised
- ▶ **Open-Source Suite – 16+ public repos** (2023 – 2026) – **mydesk** (AnyDesk-like remote control), **macscp** (Swift/AppKit SFTP), **snmp-alarm** (SNMP trap daemon), **SignalR→WS gateway**, **nginx-RTMP for macOS**, **SuperCharge SG 2026**, **Free LLM API Keys**, **Math Practice App**, 6 Redmine plugins (**forum-pdf**, **doc-pdf**, **turbo-nav**, **issue-emailer**, **dark-mode**, **custom-mail**)

- ▶ **Earlier flagship work** (2015 – 2024) – P2P Remote Desktop (C++/Qt, WebRTC, H.264/VP8); Custom Dev TUI Suite (create-project-tui, audio-tui-sox, db-exporter, debmaker); Job-Search AI TUI (9 sources, AI scoring & cover letters); n8n / Ollama self-hosted LLM infra; Telegram Dating Bot; Kids Education App (TF/Flutter); DMP crime clustering; RFID ERP + encrypted IRC for Bangladesh Army; UK e-commerce platform

EDUCATION

Doctor of Philosophy (PhD) – Computer Science <i>Lincoln University College · Part-time</i>	Jul 2025 – Ongoing
Master of Computer Science <i>Lincoln University College · Part-time – Thesis: AI-Driven Medical Assistant</i>	Sep 2023 – May 2025
BSc in Computer Science & Engineering <i>United International University · Full-time – Project: An Intelligent Building Power Management System (IoT)</i> Coursework: Java, PHP, Python, .Net, Ruby, J2EE, Embedded Systems, ML, Neural Networks, Data Mining, Statistical Inference. Awarded certifications in Programming Contest & Robotics.	2009 – 2017

CERTIFICATIONS & AWARDS

- ▶ Microsoft Imagine Cup 2012 – World’s Premier Student Tech Competition
- ▶ Programming Contest Winner – Qazi Azher Ali Memorial, 2011
- ▶ UIU Robotics Workshop Certification, 2011
- ▶ NYDSA – National Youth Development Certificate
- ▶ Code Warriors Challenge – Programming Competition

ADDITIONAL CONSULTANCY

- ▶ **Alamin Sweaters** (Feb – May 2013) – Production workflow optimization, barcode tracking
- ▶ **Friends Appeals** (Mar – Aug 2018) – Business management software, DB architecture, reporting
- ▶ **Peregrine Air Line (UK)** (Jan – Jun 2018) – E-commerce, stock management, payment gateway